

## HOUSTON EXECUTIVE, TX, USA

WMO: 720617

Lat: 29.800N

Lon: 95.900W

Elev: 168

StdP: 14.61

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 00208

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.9       | 32.0 | 16.4                          | 12.7 | 37.9 | 21.0 | 15.8 | 44.4 | 25.8                  | 58.4 | 23.9 | 56.3 | 5.4                   | 330  | 0.423 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 20.2                   | 98.9            | 76.1 | 95.5 | 76.6 | 94.6 | 76.8 | 82.7                | 84.6 | 81.4 | 84.3 | 80.6 | 85.5 | 8.4                  | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 82.2                            | 168.5 | 82.9 | 81.1 | 162.8 | 82.4 | 79.3 | 153.0 | 81.7 | 46.7          | 83.1 | 45.1 | 83.6 | 44.3 | 86.2 | 93.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 22.4              | 19.6 | 17.7 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |
|                   |      |      | 21.5                       | 82.6 | 4.8                | 1.8 | 18.0                                               | 83.9 | 15.1       | 84.9 | 12.4       | 85.9 | 8.9        | 87.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 69.6   | 52.1  | 56.9  | 62.8  | 69.3  | 76.0  | 82.0  | 83.8  | 84.6  | 80.1  | 71.0  | 60.7  | 55.1 |
|                                                                               | DBStd       | 13.50  | 9.77  | 10.50 | 8.77  | 6.62  | 5.64  | 2.95  | 2.35  | 3.06  | 4.08  | 7.33  | 10.19 | 9.90 |
|                                                                               | HDD50       | 233    | 90    | 48    | 12    | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 22    | 60   |
|                                                                               | HDD65       | 1422   | 412   | 262   | 146   | 34    | 5     | 0     | 0     | 0     | 0     | 29    | 200   | 334  |
|                                                                               | CDD50       | 7392   | 156   | 242   | 410   | 580   | 805   | 960   | 1046  | 1074  | 902   | 653   | 345   | 219  |
|                                                                               | CDD65       | 3104   | 13    | 35    | 78    | 164   | 345   | 510   | 581   | 609   | 452   | 217   | 72    | 28   |
|                                                                               | CDH74       | 31480  | 38    | 162   | 425   | 1205  | 3105  | 5496  | 6681  | 7287  | 4404  | 2060  | 480   | 137  |
|                                                                               | CDH80       | 14233  | 1     | 20    | 62    | 280   | 1147  | 2633  | 3368  | 3905  | 1985  | 730   | 93    | 9    |
| Wind                                                                          | WSAvg       | 7.5    | 8.0   | 8.9   | 8.9   | 9.2   | 8.7   | 6.6   | 5.9   | 5.8   | 5.5   | 6.4   | 7.6   | 8.0  |
| Precipitation                                                                 | PrecAvg     | 48.2   | 3.4   | 2.9   | 3.5   | 4.1   | 4.8   | 4.4   | 3.6   | 3.9   | 4.3   | 4.9   | 4.1   | 3.6  |
|                                                                               | PrecMax     | 66.9   | 9.3   | 8.9   | 8.3   | 11.8  | 12.2  | 12.8  | 11.7  | 9.3   | 12.6  | 12.1  | 14.4  | 9.6  |
|                                                                               | PrecMin     | 16.3   | 0.2   | 0.6   | 0.4   | 0.1   | 0.0   | 0.1   | 0.3   | 0.1   | 1.0   | 0.0   | 0.2   | 0.9  |
|                                                                               | PrecStd     | 12.7   | 2.2   | 2.0   | 2.1   | 3.1   | 3.2   | 2.9   | 3.1   | 2.4   | 2.7   | 3.9   | 3.1   | 1.9  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 77.4  | 82.3  | 84.2  | 89.9  | 93.4  | 99.4  | 99.1  | 101.9 | 98.8  | 91.2  | 86.1  | 81.4 |
|                                                                               |             | MCWB   | 66.7  | 69.1  | 70.2  | 70.2  | 74.3  | 75.0  | 76.1  | 75.4  | 73.7  | 75.1  | 74.9  | 70.8 |
|                                                                               | 2%          | DB     | 74.6  | 78.9  | 81.5  | 85.5  | 90.8  | 95.3  | 96.6  | 99.4  | 95.0  | 90.1  | 82.2  | 78.6 |
|                                                                               |             | MCWB   | 65.6  | 68.0  | 69.0  | 71.3  | 75.1  | 75.6  | 76.6  | 76.7  | 74.8  | 75.2  | 71.6  | 69.5 |
|                                                                               | 5%          | DB     | 72.0  | 75.1  | 79.0  | 82.4  | 89.6  | 93.1  | 94.8  | 97.2  | 92.5  | 86.4  | 80.6  | 74.7 |
|                                                                               |             | MCWB   | 65.7  | 66.6  | 68.0  | 70.5  | 74.8  | 77.0  | 77.3  | 77.1  | 75.8  | 74.0  | 70.7  | 68.9 |
|                                                                               | 10%         | DB     | 67.9  | 72.3  | 75.4  | 81.2  | 86.1  | 91.1  | 92.9  | 94.9  | 90.3  | 84.1  | 76.5  | 72.2 |
|                                                                               |             | MCWB   | 63.5  | 66.3  | 66.7  | 69.4  | 74.5  | 77.5  | 77.7  | 77.4  | 76.3  | 72.0  | 69.4  | 67.8 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 72.4  | 72.3  | 73.4  | 77.4  | 83.9  | 84.5  | 84.1  | 83.8  | 88.4  | 80.5  | 77.9  | 74.7 |
|                                                                               |             | MCDB   | 73.6  | 76.9  | 77.9  | 82.2  | 83.9  | 84.8  | 85.1  | 85.7  | 88.4  | 83.0  | 81.3  | 76.9 |
|                                                                               | 2%          | WB     | 69.3  | 71.1  | 72.2  | 75.2  | 78.6  | 81.4  | 81.8  | 82.0  | 82.0  | 78.0  | 75.4  | 72.7 |
|                                                                               |             | MCDB   | 71.7  | 75.8  | 77.4  | 80.6  | 84.6  | 84.9  | 84.4  | 84.9  | 83.5  | 83.6  | 78.9  | 75.4 |
|                                                                               | 5%          | WB     | 67.2  | 69.9  | 70.9  | 73.9  | 77.5  | 80.5  | 80.9  | 81.1  | 80.2  | 76.9  | 73.6  | 71.0 |
|                                                                               |             | MCDB   | 70.6  | 73.5  | 76.1  | 78.6  | 83.2  | 85.7  | 85.3  | 85.4  | 84.8  | 82.8  | 76.8  | 74.5 |
|                                                                               | 10%         | WB     | 64.3  | 67.9  | 69.6  | 72.8  | 76.6  | 79.1  | 79.7  | 80.1  | 78.7  | 75.5  | 72.0  | 68.5 |
|                                                                               |             | MCDB   | 65.9  | 71.2  | 73.8  | 77.8  | 82.2  | 86.1  | 87.4  | 87.0  | 84.7  | 81.3  | 75.3  | 71.3 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 20.7  | 19.2  | 20.3  | 20.0  | 18.5  | 18.3  | 18.7  | 20.2  | 19.1  | 22.7  | 21.7  | 19.8 |
|                                                                               |             | MCDBR  | 21.8  | 21.1  | 20.3  | 20.7  | 19.7  | 20.8  | 21.0  | 23.5  | 22.0  | 22.1  | 21.2  | 22.0 |
|                                                                               |             | MCWBR  | 14.6  | 12.1  | 10.4  | 10.0  | 6.4   | 6.5   | 6.5   | 6.8   | 8.4   | 10.1  | 11.8  | 15.5 |
|                                                                               | 5% WB       | MCDBR  | 18.4  | 17.3  | 18.0  | 16.9  | 16.2  | 17.3  | 17.8  | 20.5  | 17.5  | 19.3  | 17.8  | 20.2 |
| MCWBR                                                                         |             | 14.5   | 12.2  | 12.1  | 10.4  | 7.6   | 7.9   | 7.3   | 7.1   | 9.6   | 10.5  | 12.2  | 17.0  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.339  | 0.356 | 0.373 | 0.433 | 0.480 | 0.463 | 0.490 | 0.479 | 0.448 | 0.380 | 0.373 | 0.343 |      |
|                                                                               | taud        | 2.483  | 2.451 | 2.394 | 2.232 | 2.139 | 2.242 | 2.186 | 2.218 | 2.269 | 2.444 | 2.407 | 2.485 |      |
|                                                                               | Ebn at Noon | 284    | 288   | 289   | 274   | 261   | 264   | 256   | 258   | 263   | 276   | 268   | 276   |      |
|                                                                               | Edh at Noon | 29     | 33    | 37    | 45    | 49    | 44    | 47    | 45    | 41    | 33    | 31    | 28    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 911    | 1029  | 1396  | 1676  | 1853  | 1963  | 1950  | 1859  | 1591  | 1366  | 1025  | 809   |      |
|                                                                               | RadStd      | 89     | 185   | 122   | 125   | 135   | 153   | 174   | 134   | 160   | 165   | 109   | 100   |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (36 neighbors) | +0.58 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | +189  | +167  |       |

Nomenclature: See separate page